

Weekly Status Report

Name	Week / Date
Craig Rudman	Week 6 / 2026-08-09

Project title

Predicting Region-Season Wildfire Cause Patterns to Target Prevention and Mitigation

Project summary

This project helps a state or regional fire planner match prevention and mitigation effort to the pattern that actually drives burned acres, instead of spreading it uniformly. The data is the Fire Program Analysis Fire-Occurrence Database (FPA-FOD), about 2.3M U.S. wildfires from 1992 to 2020, grouped by cause, EPA Level III ecoregion, and meteorological season.

The model works in two steps. First it splits a region-season's burned acres across three classes — Human, Natural, and Unknown, where Unknown means no cause was ever recorded. Then each class gets its own follow-on model, because each raises a different question.

The predictive deliverable comes from the first step plus the Human class, the only class with a composition of causes to rank. For a region and upcoming season it ranks causes by acres likely to burn rather than by fires started, so effort goes where the acres are. The ranking is the reliable part: the cause mix is stable and predictable from history, while the season's total burn is not, so the acre figures carry much wider error than the order does.

Natural is the largest first-step class at 58.9% of acres, and its branch changed this week. It previously asked where lightning-driven acres would concentrate; it now asks where within a region fires are most likely to start, on a fine hexagonal grid rather than the whole ecoregion — because fuel treatment, defensible space and pre-positioned crews are all sited works, and whether they pay off depends on whether fire arrives there. The two views disagree usefully: ignitions and acres concentrate in different places, so a cell can run high-ignition/low-acre (starts caught early) or low-ignition/high-acre (rare starts that run far), and those call for different treatments. The pattern is consistent with ignitions clustering near roads and settlement while acres accumulate in remote continuous fuel, though no road or settlement layer was built to test that.

Unknown is not a forecast but a data-quality signal: where a region's causes go unrecorded, the record itself is the weak link. The branch ranks the regions and reporting streams where better cause data would most improve everything above it.

One result recurs across the project and is worth stating separately, because building the storyboard is what made it visible as a single thing. Every failure this project has recorded is a **grain mismatch**: W4's climate covariates carried a real per-region signal that averaged to zero when pooled across 105 regions; the hex-grain covariates identify dry *places* rather than dry *years*; the ecoregion turned out too coarse to site anything, which the hex fixed; and burned-area size is unforecastable over a season, which the same logic suggests may need the day. Three of those were diagnosed separately before the fourth made the

pattern obvious. The practical form of it is that the model's unit has to match the unit the phenomenon actually varies on — and where it does, cheap history beats sourced covariates every time.

The research questions

The rest of this report refers back to these:

- RQ1 (descriptive): Across a set of contrasting U.S. region-seasons, which wildfire causes (natural and human) drive the most burned area, and do those patterns differ enough to demand different prevention and mitigation strategies?
- RQ2 (predictive): Can a next-season cause-risk profile — the expected composition of causes for a region and upcoming season, ranked by the burned area each is expected to drive — be predicted well enough to pre-target that prevention effort?

Milestones

- **Done** — Data acquisition, feasibility, and EDA: FPA-FOD 1992–2020 loaded and validated; cause composition, size distributions, and the differential missing-cause problem characterized.
- **Done** — Cleaning: documented exclusions; EPA Level III spatial join (CONUS and Alaska); meteorological season and season-year index derived.
- **Done** — Method design: hierarchical structure settled over a flat classifier, with the estimator choice defended against the alternative.
- **Done** — Perimeter correction and hex grid (W5): MTBS perimeters joined, acres distributed across res-5 hexes, conservation verified at 99.61% on-grid.
- **Done** — Feature engineering: trailing cause and burn history; pre-season climate at both region and hex grain; prior-burn state per hex-season; MODIS vegetation density, which cleared the credential blocker carried since W5. All external features lagged to pre-season availability.
- **Done** — Level III modeling: persistence baselines for all three branches, forward-chaining splits, and learned rungs on Tier 1 and Human. The k=7 trailing mean stands as the model to beat.
- **Done** — Hex-grain modeling (W6): ignition and burned-area targets for the Natural and Human branches, with shuffled-control tests and covariate ablation ladders.
- **Done** — Findings and the storyboard: nineteen assertion headlines in delivery order, each with a figure, and the recommendation reduced to one sentence the deck opens and closes on.
- **In progress** — The final deck: laying the nineteen figures into slides, and the W6 practice recording against the five-minute subset.

Last week's "To Do"

- Source the pre-season fuel-condition and climate layers. **Done** — climate at hex grain ([src/hex_climate.py](#)), and MODIS vegetation density ([src/hex_ndvi.py](#)), 55,923 hex-seasons across six forest ecoregions.
- Add a learned rung to the Human branch. **Done, and it lost** — acre-weighted TVD 0.489 → 0.588, top-1 54% → 36%, against the k=7 persistence floor.
- Enforce the leakage rule as the imagery features are built. **Done** — every covariate ends strictly before the target season opens, with anomalies normalized on training years only.

This week's progress

BLUF — the one-sentence recommendation

Stop targeting how big it gets. Target where it starts. Rank causes by region, site the work by ignition, and fix the record where it says neither — the order is trustworthy, the acre level much less so.

Reworded this week to lead with the change being asked for rather than with the three products that deliver it. The previous version — *"target prevention where the region's own history says the causes are, site mitigation where it says fires start, and fix the record where it can't say either"* — is accurate and describes the deliverables, but a planner reading only the first line should learn what to do **differently**, and that is the shift away from forecasting fire size. Beat 19 closes on the same sentence, so the deck's first and last statements match.

Deliberately a targeting claim, not an efficacy one. Nothing in this project measures what a treatment achieves; the ranking says where fire is most likely to arrive, which is a necessary condition for sited work to pay off and not a sufficient one.

Storyboard — assertion headlines in order

Each headline is a full-sentence assertion, and the sequence is meant to carry the argument with the slides removed. The evidence column names what the slide would show, not what it would say.

#	Assertion headline	Evidence on the slide
1	Wildfires are seasonal.	img/w6_seasonality.png — starts and acres on one calendar, no axes and no magnitudes: most fires start in spring, most acres burn in summer. MAM and JJA start nearly the same number of fires and differ 3.9x in acres.
2	Cause is regional, not national.	img/w6_cause_map.png — all 105 Level III ecoregions shaded by natural share of attributed acres. A West/East split at roughly the 100th meridian, Alaska almost entirely natural. The distribution is bimodal: 50 regions below 20% natural, 28 above 80%, only 27 in between.
3	A region's cause mix is stable enough to forecast.	img/w6_tier1_tiles.png — three tiles worst to best, share of a region-season's burned-acre composition placed on the right cause: the national average mix 42%, an even split 52%, the region's own history 73% . Acre-weighted TVD 0.580 / 0.485 / 0.266, forward-chained on 2010+, 3,949 held-out region-seasons.
4	A few years of rolling average is enough.	img/w6_k_sweep.png — error in the predicted cause mix against window length. Acre-weighted TVD falls 0.331 → 0.278 from one prior season to three, then flattens; every window from three up sits within 1.4 points of the best. Cause composition is a standing property, not a yearly swing, and the window length is not a decision worth agonizing over.
5	Within Human, history names the lead cause	img/w6_human_tiles.png — the same three tiles as beat 3, one level deeper: how often the predicted leading human sub-cause is the right one out of 11. An even split 9%, the national human mix 16%, the region's own

#	Assertion headline	Evidence on the slide
	more often than not.	history 54% . Acre-weighted TVD 0.489 against the national mix's 0.643, forward-chained on 2010+, 3,850 held-out region-seasons.
6	A learned model made it worse.	img/w6_human_ladder.png — three rungs against the floor line. The region's own history names the leading human cause 54% of the time; gradient boosting on region character 36%; the same model <i>given that history as a feature</i> 47%. Acre-weighted TVD 0.489 / 0.588 / 0.554. Even handed the winning quantity, the model cannot beat taking its mean.
7	Where fires start is predictable, but not at ecoregion scale.	img/w6_siting_glance.png — Klamath hexes in two bands, labeled on the map with no legend: the deep band is 6% of the region catching 32% of next season's starts (5.2x), the light band 29% for 60% (2.1x). The bands are the map's version of the capture curve's first two marks, so the geography of the diminishing return is visible. Held-out Spearman: Human +0.53, Natural +0.34. Companion for Q&A — img/w6_capture_curve.png , which carries the decay out to 1.16x at 90%. This headline also carries the deck's one grain change — the ecoregion of beats 3–6 becomes a 62,494-acre hex, and the map is the first thing the audience sees at that scale.
8	That skill is spatial, not statistical luck.	img/w6_shuffled_control.png — one panel, two lines labeled forecast and shuffled : mean observed starts against mean predicted for 20 equal-count strata. The forecast climbs the diagonal (top stratum predicted 4.7, observed 3.7); the same predictions dealt to the wrong hexes go flat at ~0.4 whatever the prediction. Spearman +0.526 → +0.0002 , MAE 0.43 → 0.77 (worse than the uniform baseline's 0.70), on 1.59M held-out hex-seasons.
9	Human fire is predictable year-round; lightning only in summer.	img/w6_season_skill.png — held-out Spearman by season, each branch scored separately in all 11 held-out years , with the band spanning the observed year-to-year range. Human runs flat and high (median +0.47 to +0.61, peaking in spring); natural is a summer surface (+0.42 JJA, +0.07 DJF). Human beats natural in all 44 season-years without exception. The deck's only figure showing a distribution rather than a point estimate.
10	Almost all the acres are in almost none of the cells.	img/w6_acres_concentration.png — cumulative share of natural acres, cells ordered by acres burned, least to most . The curve hugs the floor and spikes at the right edge: the worst-burning 10% of cells hold 98% of the acres, the worst-burning 1% hold 55% . On its own this is a fact about how fire is distributed; it sets the stakes for the next slide by naming, before any error is shown, which cells a forecast has to get right.
11	Up to a point, both are predictable. Past that point, natural fire is harder.	img/w6_branch_deciles.png — typical forecast error against how much a cell burned, both branches on one axis, the same axis and the same direction as beat 10 , so the right edge means "the big burns" on both slides. They track each other through the smallest third and separate steadily after: natural runs 2–3x worse than human at the same cell size , reaching 687x on a median 8,061-acre cell against human's 19x on 240. The separation lands exactly on the right edge beat 10 just marked. Cells under 1 acre are excluded

#	Assertion headline	Evidence on the slide
		— 25.3% of FPA-FOD rows sit at exactly 0.1 acres (44.5% of natural fires), a reporting default rather than a measurement.
12	We tried to fix it with drought and fuel, but where fires start is a property of the place, not of the year.	img/w6_ignition_ladder.png — two flat lines, one per branch, across the rungs added to improve them: the region's own history, + drought, + fuel load, + both. Nothing moves. Best gain on either branch is +0.0045 ; the y-axis runs from zero so a real effect would have been visible. Both branches on the same 29,293 held-out cells (JJA, the six forest ecoregions NDVI was fetched for), so the two lines are directly comparable. The opening of the repair attempt, and it has to be said as one — beat 11 leaves a failure on the table; beats 12–14 are what was done about it, not a new topic. The measured reason is spoken, not drawn: these layers identify dry <i>places</i> , not dry <i>years</i> (raw vs. within-hex anomaly, pdsi $-0.137 \rightarrow -0.073$, NDVI $+0.228 \rightarrow +0.098$), and place is what history already knows. Held for Q&A — img/w6_ndvi_variance.png , the place-vs-year split at 2.8x, which is the mechanism behind the flat line.
13	The same data does predict how much burns.	img/w6_acres_ladder.png — deliberately the same figure as beat 12: same rungs, same axis, same zero-based scale, one line instead of two. The shape is the argument. Three rungs sit on the baseline rule and the fourth leaves it: climate + NDVI together +0.049, 26.6 SD above a covariate-shuffled control, holding across five forward-chaining split years ($+0.012$ to $+0.066$). Neither half works alone — drought alone -0.008 , fuel alone +0.001 — and the figure's one callout is the size of the step, +0.049 , since the + both tick already names the conjunction: wet heavy fuel will not carry fire, dry bare ground has nothing to burn. Different target and population from beat 12 (7,799 burning JJA cells, burn-conditional baseline), so the two are comparable in shape, not cell for cell .
14	But the gain lands on the fires nobody needed predicted.	img/w6_gain_landing.png — beat 11's axis again , cells ordered by acres burned least to most, with the covariate model laid over the baseline. The shaded lens is the trade: it opens in deciles 6–9 (1–200 acre cells), and it closes at the right edge where the two lines finish together, 855x $\rightarrow$ 868x on a median 5,073-acre cell. Deciles 1–5 get materially worse — decile 1 nearly doubles, 18.8x $\rightarrow$ 35.9x — so the model buys middling cells by giving up small ones and gains nothing where it counts. Beat 10 is what makes this fatal rather than disappointing: the improvement misses the cells holding 98% of the acres. Third beat to land the eye on the same right edge, which is the argument for reframing rather than tuning. 780 held-out cells per decile.
15	Siting needed a finer place. Size needs a finer moment.	img/w6_grain_parallel.png — the deck's only figure that plots no data, and its only one about the method rather than the fire. Every number this beat could show has already been shown (the tail is beat 11, the concentration beat 10, the nulls 12–14), and re-plotting any of them invites the audience to re-audit numbers at the moment the talk needs them accepting a conclusion. What it draws instead is a parallel: beat 7 fixed a

#	Assertion headline	Evidence on the slide
		forecast by dropping grain in <i>space</i> — the region was too coarse to site anything, so the map broke into hexes — and the tail failure is the same defect on the <i>time</i> axis, where a season is too coarse to say how big a fire gets. The solved row is closed and the untested row dashed, because one drop is a result and the other a hypothesis. This is the fourth time the project has met the same lesson (W4's pooled climate null, beat 12's places-not-years, beat 7's grain drop, now this) and the first time it is named as one. Megafire size remains not forecastable <i>before the season</i> — the bounded claim — but the reason is now stated as a grain mismatch rather than a dead end, which is what lets beat 16 move upstream instead of consoling.
16	Every megafire was an ignition first.	img/w6_ignition_gate.png — two bars in a common frame, no axis: a hex-season that ignites at all produces a $\geq 1,000$ -acre burn 6.7% of the time against 0.29% for one that does not, a 22.8x gate. JJA natural, held-out years. The deck's only ratio-between-two-probabilities, so it uses the plainest encoding available rather than reusing a learned axis; the percentages are printed because at true scale the 0.29% bar is nearly invisible — which is itself the finding. This is the door beat 15 left open: size is not forecastable, but the event upstream of it is, and beat 7 already showed that ignition location is the most predictable surface in the project.
17	One ignition is enough.	img/w6_one_is_enough.png — one stacked bar over all 2,724 large-fire cells in the held-out years, split by how many times their hex ignited that season: 49% had exactly one , 21% two, 30% three or more. Only the first segment carries color, because it is the ground a planner would deprioritise by ranking on ignition count. Held for Q&A — the rate curves showing escape probability <i>per ignition</i> falling $0.054 \rightarrow 0.014$ as count rises, which answers "but doesn't more ignitions mean more risk?" without putting a rate-versus-total subtlety on the slide. The rule is binary: does this place ignite, not how often.
18	Nearly a fifth of burned acres have no specific cause — and that gap is itself forecastable.	img/w6_unknown_triage.png — the ranked triage list, headed by Southwestern Tablelands MAM at 1.17M predicted unattributed acres. Ranked by acres rather than by rate: Central Great Plains has the worse attribution rate (66%) but a fifth of the burn. Unknown-branch persistence: acre-weighted MAE 0.167 against the global mean's 0.240, so a planner can be told a season ahead where next year's record will fail. This is the third leg of the recommendation, delivered where it is actionable rather than as a finding in its own right.
19	Target causes by region. Site the work by ignition. Fix the record where it says neither.	img/w6_recommendation.png — three rows, one per Tier-1 class, each with its share of acres, its instruction and its grain: Natural 58.9% , site by ignition, hex-season; Human 22.7% , rank causes by the acres they drive, ecoregion-season; Unknown 18.5% , fix the record underneath, ecoregion-season. Ordered by share rather than by the headline's clause order, so the top row is where the acres are. The Tier-1 split from beat 3 returns with a

#	Assertion headline	Evidence on the slide
		deliverable on each class, which makes the three-part recommendation structural rather than three unrelated tips. The slide closes on the recommendation itself — "Stop targeting how big it gets. Target where it starts" — which recovers the instruction that came off beat 15's headline, with its boundary on the line beneath: rank on all three, the order is trustworthy and the acre level much less so. A statement rather than a chart; see below.

The five-minute cut, for the W6 practice recording

The final is 15 minutes — about 10 to present, 5 for questions — which is what the nineteen beats above are sized for, at roughly 30 seconds each. The W6 practice talk is 5 minutes, so it is a **subset of the same storyboard rather than a second one**: beats **1, 2, 3, 7, 10, 15, 16, 17, 19**.

That keeps the spine intact — fire is seasonal, cause is regional, the mix is forecastable, starts are predictable at hex grain, the acres sit in almost no cells, the grain diagnosis, the gate, the binary rule, the recommendation. What it drops is every beat whose job is to *defend* a result rather than state one: the k-sweep (4), the losing learned rung (6), the shuffled controls (8), the four-season human profile (9), the two-branch tail split (11), and the whole covariate arc (12–14). Those are the beats most likely to draw questions, which is exactly why they stay in the final and come out of the rehearsal.

Two beats drop for a different reason. Beat 5 (the Human sub-cause floor) is a result rather than a defense, but the five-minute cut carries only one level of the hierarchy, and beat 3's tiles teach the encoding that beat 5 reuses — so the deeper level goes and the shallower one stays. Beat 18 (Unknown) is a third of the recommendation and survives as a row on beat 19, but at five minutes a track that appears once cannot carry its own slide.

What the cut costs, stated plainly. Without 11–14 the five-minute version asserts that acres concentrate (10) and then jumps to the grain diagnosis (15) without ever showing the forecast failing on the tail or the covariates failing to fix it. Beat 15 has to carry that on the strength of the speaker's word. That is acceptable in a rehearsal and would not be in the final, which is the reason the full nineteen exist.

Headline-only test

Read as a sequence with the slides removed, the nineteen assertions carry the argument: fire is seasonal and cause is regional (1–2), what to target and why history suffices (3–6), where to site and why the skill is real (7–9), where the acres actually sit and where the branches diverge (10–11), what fuel data can and cannot buy (12–14), the grain diagnosis that bounds the null (15), the reframe that turns it into a recommendation (16–17), where the record itself is the weak link (18), and the three-part recommendation (19).

The test is self-administered so far, which is its weakness. The assignment runs it with a podmate reading only the headlines, and that is the version that finds gaps — an author reading their own headlines supplies from memory the connective tissue a listener does not have. Everything below is what the test surfaced from the inside; the pod-feedback list names what I most need an outside reader to check.

The descent, and what carries a listener through it. Beats 10 to 15 are six consecutive statements of what does not work, which is the longest stretch in the deck without a positive result. Three things keep it from reading as a list of disappointments. The run opens on a fact about fire rather than a forecast failure — almost all the acres are in almost none of the cells — so the stakes are set before anything fails. Beat 12's headline frames the covariate arc as a repair attempt, so 12–14 are *what we did about it* rather than three more losses. And beat 15 ends the descent by naming the failure's shape instead of restating it, which is what lets beat 16 move upstream. Whether that is enough is the second question in the pod-feedback list below.

The seam between the two grains, and where it is announced. Beats 3–6 and beats 7–9 are different sub-projects: the grain changes from 105 ecoregions to 36,234 hexes, the target from acre composition to ignition counts, and the unit from shares to counts. Three changes at one boundary, and a listener has no reason to expect any of them. Beat 7's headline carries the grain change — *"but not at ecoregion scale"* names what is being left behind rather than only what is arriving. The target change is not something a headline can hold, so it is a spoken line in [talk_notes.md](#): this beat predicts **where fires start**, not how much burns, and the two are not the same model. Left unsaid, the capture curve reads as "32% of the *burn* under 6% of the ground," which is a stronger claim than the evidence supports.

The second seam, at 11→12. The same defect in a different form. Beat 11 ends on a failure and beat 12 opens on drought and vegetation indices, which invites reading the covariate arc as a change of subject when it is the *response* to the failure just shown. Nobody sourced climate and NDVI out of curiosity — they were sourced to fix exactly what beat 11 exposes. Beat 12's headline says so outright (*"we tried to fix it with drought and fuel, but..."*), which makes 12–14 a repair attempt with a verdict rather than three unexplained slides of covariate results. It also changes what the null *means*: a covariate that fails after being introduced as a fix is evidence about the problem, where the same covariate failing in a vacuum is just a dead end.

The two slides that show no data, and why. Beats 15 and 19 are the only figures in the deck that plot nothing, and both earn it the same way: what they need to show is not a measurement. Beat 15 draws the *scope* of a null — the grain drop that worked beside the one that was never tested — because every number it could plot had already been plotted three beats earlier, and re-plotting it would invite the audience to re-audit numbers at the moment the talk needs them accepting a conclusion. Beat 19 draws the *structure* of the recommendation, because its headline is three instructions pointing at three products at two grains: any single chart has to pick one and demote the other two, and each candidate — the siting map, the triage list, the Tier-1 tiles — is already a slide the audience has seen. What had never been shown is all three together, which is the one thing that makes the recommendation read as architecture rather than as three unrelated tips.

The cost is that a deck partly judged on visuals now ends with a no-data slide. Three beats separate them, and beats 16–18 are all figures, so the run does not read as a deck that ran out of material.

The buried lead, and where it moved. The first storyboard was twelve beats in chronological order — build the hex grid, fetch the covariates, run the ablations, discover the tail failure, then notice the ignition gate — which buried the lead at **position 11 of 12**. It had a second and larger problem: it opened on the hex natural surface, which is *one of three tracks* in a two-tier architecture, and presented it as though it were the whole product. RQ2's actual deliverable, the region-season cause profile, appeared nowhere. The rewrite restored the hierarchy — Tier 1 establishes **what**, hex grain answers **where**, the

recommendation joins them — and the ignition gate now arrives as **beat 16 of 19**, where it opens the recovery rather than closing the deck.

Two further moves followed. Neither was a buried lead; both were beats sitting where they could not do their work.

what moved	from → to	why
the Unknown branch	7 → 18	It is a recommendation about the record, not a finding, and at beat 7 it interrupted the Human thread to deliver something a planner could not act on for another eleven beats. Against beat 19 it becomes the third of three Tier-1 products, all on screen together. The cost: Tier 1 introduces a class at beat 3 that nothing exercises until 18, so beat 19 reintroduces it in a sentence.
the concentration curve and the tail failure	11, 10 → 10, 11	The stakes have to precede the failure. Naming that almost all the acres sit in almost none of the cells <i>before</i> showing the forecast break there makes beat 11 a payoff instead of one more disappointment.

The through-line in all three is the same: chronology is a fact about the project, and the audience does not care in what order things were tried.

Why the concession and the reframe are adjacent. Beat 15 and beat 16 are the deck's most fragile join — a listener has no reason to expect that "size is not forecastable before the season" leads to "every megafire was an ignition first," and read carelessly they are a failure followed by an unrelated consolation. Beat 15's headline is a *diagnosis* rather than a bare null for exactly this reason: naming the failure as a grain mismatch points somewhere, and where it points is upstream, at the ignition. Beat 11 carries a second load in the same service — showing that only lightning fire has a runaway tail is what licenses beat 19 to ship two *different* products, so it cannot be compressed away.

Why the covariate arc is three beats and not one. Reported as a single null it is a disappointment. Split, it is a contrast: the same fuel data that says nothing about *where a fire starts* (12) does add real signal about *how much burns* (13, +0.049 at 26.6 SD above a shuffled control), and then lands on 1–20 acre fires while the tail goes 855x to 868x (14). That contrast is what explains why the recommendation sites against starts rather than against predicted burn size.

What still needs pod feedback. Two questions, in order of how much they would change.

First, whether the deck reads as one argument or as two stapled together — the Level III *what* and the hex-grain *where*. Beat 7's headline and a spoken line in [talk_notes.md](#) carry the join; whether they are enough is exactly what an outside listener can tell me and I cannot.

Second, whether beats 10–15 read as a descent. Six consecutive negative results is the longest stretch in the deck without a positive one, and each is load-bearing: without 11 the two branches are not distinguished, and without 12–14 beat 15's diagnosis is unsupported. If a podmate loses interest anywhere in that run, the ordering is not carrying it and compression is the answer.

The compression candidate if it runs long is still beat 4 — it tunes a parameter of the beat-3 baseline rather than adding a claim, and its finding is that the parameter barely matters. Beat 6 (the learned rung

losing) reads as the same "the simple thing won" note but earns its place: it is the only beat that concedes a model was tried and lost, which is what keeps the nulls credible.

Built this week

- `src/hex_ignitions.py` — ignition-count target per hex-season, 4,166,910 cells, all 2.27M fires on raw points (no perimeter join: an ignition is exactly what the record stores correctly).
- `src/hex_acres.py` — burned-area target with a hurdle split (does it burn / how much), plus the burn-conditional baseline that corrected a reversed finding.
- `src/hex_panel.py` — one panel assembly, one persistence baseline, one scorer. Calls `trailing.TrailingMean` rather than reimplementing it; results are cached so the floor is quoted rather than recomputed.
- `src/hex_climate.py`, `src/hex_ndvi.py` — pre-season covariates at hex grain. The MODIS probe reaches the Planetary Computer STAC API anonymously, which cleared the Earthdata/GEE credential blocker carried since W5.
- `notebook/12_hex_ignition_baselines.ipynb`, `notebook/13_hex_acres_baselines.ipynb` — the two natural-branch targets, their baselines, the shuffled-control tests, and the ablations.
- `notebook/14_hex_human_branch.ipynb` — the Human branch at hex grain. No new data was needed; the targets were already in the cached panels. Scored per season, because human fire runs in all four where natural ignition concentrates 78.1% in summer.
- `src/w6_visuals.py`, `notebook/15_w6_visuals.ipynb` — **the deck's figures, one per storyboard beat, nineteen of them.** Every number a figure prints is either computed from a cached panel or transcribed from the notebook that published it, with a module-level source constant naming the cell — so a figure cannot drift from the report that quotes it. Four decisions govern the set:
 - **Reuse an axis rather than teach a new one.** Beat 11's "cells ordered by acres burned, least to most" appears three times (10, 11, 14), so beats 11 and 14 spend their attention on the finding rather than on the frame. The Tier-1 tiles (3) and the Human tiles (5) share `_draw_score_tiles` for the same reason: one question at two levels should not be two encodings.
 - **Draw the null on a scale where a real effect would have shown.** The ignition ladder (12) runs from zero, so that four flat rungs read as flat rather than as noise magnified by a cropped axis; the acres ladder (13) uses the same zero-based scale for the opposite reason, so its one real gain is not inflated by the axis into something beat 14 then has to deflate.
 - **Two slides plot no data**, because what they show is not a measurement: beat 15's grain parallel and beat 19's three products. Both are argued above.
 - **Populations are named wherever two figures could be mistaken for one comparison.** The ignition and acres ladders look deliberately alike and are scored on different targets, baselines and cell counts (29,293 hex-seasons against 7,799 burning cells); the storyboard says "comparable in shape, not cell for cell" so the resemblance cannot be misread.

The two figures held for Q&A rather than the deck are the NDVI variance decomposition (between-place spread 2.8x the within-hex spread, which is the *mechanism* behind beat 12's flat line) and the region volatility map. Both answer questions well and neither survives the glance test as a slide.

Next week's "To Do"

- **Fix the point attribution of large unperimetered fires.** Found this week: a point-only fire puts its whole acreage on the single hex holding its ignition, which is right at the 14-acre average and wrong in the tail — **2,710 point fires over 1,000 acres carry 8.9% of all acres**, and 23 assign more than a full hex to one cell, including the record's 606,945-acre maximum at 971% of a hex. The fix is to impute a circular burn of the correct area from the ignition point and distribute it with the same weight machinery `hex_burn` applies to perimeters; a 606,945-acre circle has a 28 km radius against a hex's 9.9 km, so it spreads across ~10 cells. Two things to carry into the build: fires elongate along wind and terrain, so a circle errs directionally in a product that sites work by location, and rebuilding `hex_acres_res5.parquet` invalidates notebooks 13–15 and every acres figure in the deck. Rank statistics mean the W6 findings do not move, which is why this is W7 work rather than a W6 correction.
- **Record the five-minute practice talk and post it with the self-assessment.** The cut is specified above — beats 1, 2, 3, 7, 10, 15, 16, 17, 19 — and every one of those beats has its figure, so what is missing is slides to record against and the recording itself.
- **Run the headline-only test with a podmate reading, and complete the structure peer-review.** The test in this report is self-administered, which is the weakest form of it: the author cannot hear the gaps the headlines leave. Two podmate drafts also need a buried lead found and relocated.
- **Build the final deck against this storyboard, one slide per assertion.** All nineteen beats have a figure, rendered from `notebook/15_w6_visuals.ipynb`, so what remains is layout rather than construction. Two constraints to respect while laying them in: beat 4 is the compression candidate if it runs long, and beats 10 and 11 are a matched pair — 10 sets the stakes, 11 is the finding that lands on them — so neither survives alone.
- **Test whether same-day conditions — wind, timing, suppression availability — predict which igniting natural hexes escape.** Beat 15 states this as a grain hypothesis rather than a bare null: the season is too coarse a unit to size a fire, exactly as the ecoregion was too coarse to site one. That makes it the project's clearest next experiment and also the most expensive — a different data requirement, not a tuning exercise on the current panel.
- Optionally, a hyperparameter search on the Human branch's learned rungs. Both rungs are built and both lost (`08_human_cause.ipynb`): the coarse fingerprint scores 0.588 TVD / 35.7% top-1 and the history-aware rung 0.554 / 47.5%, against the k=7 floor's 0.489 / 54.1%. Neither was tuned, so a search is the one genuinely untried thing left on that branch.

Resources (optional)

- W6 assignment: [coursework/W6/assignment.md](#)
- W5 status report: [coursework/W5/MSDS696_W5_Status_Report.md](#)
- Short, K. C. (2022). *Spatial wildfire occurrence data for the United States, 1992–2020* (6th ed.). USDA Forest Service Research Data Archive. <https://doi.org/10.2737/RDS-2013-0009.6>
- Eidenshink, J., Schwind, B., Brewer, K., Zhu, Z., Quayle, B., & Howard, S. (2007). A project for monitoring trends in burn severity. *Fire Ecology*, 3(1), 3–21. (MTBS burn perimeters.)
- U.S. EPA. *Level III Ecoregions of the Conterminous United States; Level III Ecoregions of Alaska*.